

Appendix B: Worker efficiency and sorting quality parameters

This appendix documents the worker efficiency and sorting quality parameters used in the dynamic humanoid robot break-even simulation. Publicly reported benchmark values are presented where available. When no public worker level value is available, the public value field is reported as N/A and the model value is treated as a scenario assumption.

Table B1: Summary of base worker efficiency inputs

pph = parcels per hour. Effective parcels per shift include effective working time and quality adjustment used in the simulation workbook.

Carrier	Base pph	Shift hours	Effective time	Peak loss	Effective parcels/shift	Basis
SF Express	500	10.5	80%	15%	3,213	Base model uses public 500 parcels/hour benchmark and recruitment based shift evidence.
JD Logistics	400	10.0	80%	15%	2,448	Base model uses lower worker throughput due to lack of public JD worker level sorting evidence.

Table B2: Indicator level evidence classification

N/A means that no public worker level value was available in the collected sources.

ID	Category	Variable	Carrier/context	Scope	Public evidence	Data status
SF_MANUAL_PPH	Productivity	Parcels processed per hour	SF Express / manual sorting	Worker level comparison benchmark	Public comparison benchmark: manual sorting about 500 parcels/hour.	Public benchmark - media/non-official KPI
SF_AUTO_HUB_PPH	Productivity	Automated system throughput	SF South China operating hub	System/equipment level	Public system benchmark: automated sorting equipment processes at least 71,000 parcels/hour.	Public system level data
SF_AUTOMATION_INVEST	Peak/automation investment	Transfer hub automation investment	SF Express 2025H1	Company level operating data	Public company material: transfer efficiency +12.5%; >120 transfer automation units; >380 AGV forklifts; automation capacity >5.57 million parcels/day.	Public company material
IND_HZ_DAILY	Productivity/quality	Large logistics park daily capacity	Hangzhou Chengnan smart logistics park (industry benchmark)	Site/park level	Public industry benchmark: logistics park processes 2.2 million parcels/day with 99.99% sorting accuracy.	Public system level benchmark - company not named
IND_HZ_ACC	Quality	Automated sorting accuracy	Hangzhou Chengnan smart logistics park (industry benchmark)	Site/park level	Public industry benchmark: automated sorting accuracy 99.99%.	Public system level benchmark - company not named
SF_SHIFT	Shift	Shift length	SF Express / recruitment based sorting sample	Recruitment/job-posting level	Recruitment evidence: 5h, 10h, 10.5h, 11h, 12h and 12.5h shifts appear in postings.	Public recruitment information - verification required
SF_EFFECTIVE_TIME	Productivity	Effective working time ratio	SF Express / manual sorting	Model assumption	N/A (no public worker level effective working time ratio).	Model assumption
SF_WRONG_SORT	Quality	Manual wrong-sorting rate	SF Express / manual sorting	Model assumption	N/A (no public worker level wrong-sorting rate).	Model assumption
SF_MISSCAN	Quality	Manual missed-scan rate	SF Express / manual sorting	Model assumption	N/A (no public worker level missed-scan rate).	Model assumption
SF_DAMAGE	Quality	Damage rate	SF Express / manual sorting	Model assumption	N/A (no public worker level damage rate).	Model assumption
SF_REWORK	Quality	Rework rate	SF Express / manual sorting	Model assumption	N/A (no public worker level rework rate).	Model assumption
SF_PEAK	Peak	Peak-period human productivity loss	SF Express / manual sorting	Model assumption	N/A (no public worker level peak productivity loss).	Model assumption
SF_NIGHT	Night shift	Night shift productivity loss	SF Express / manual sorting	Model assumption	N/A (no public worker level night shift productivity loss).	Model assumption
JD_AUTO_KUNSHAN	Productivity	Unmanned sorting centre throughput	JD Kunshan unmanned sorting centre	System/equipment level	N/A	Public system level data
JD_AUTO_ASIA_DAILY	Productivity/quality	Kunshan Asia No.1 automated sorting centre	JD Kunshan Asia No.1 sorting centre	Site/system level	N/A	Public system level data
JD_SORT_WALL_PPH	Productivity/quality	Automatic sorting wall throughput	JD Logistics automatic sorting wall	Equipment level	N/A	Public equipment level data
JD_PICKER_ASSIST	Productivity	Robot-assisted picking productivity	JD Logistics robot-assisted picking	Human-robot picking	N/A	Public human-robot collaboration data
JD_SMART_WAREHOUSE	Productivity/cost	Zhilang goods-to-person system productivity	JD Logistics smart warehousing	System/equipment level	N/A	Public system level data
JD_WORKER_PPH	Productivity	Parcels processed per hour	JD Logistics / manual sorting	Model assumption	N/A	Model assumption
JD_SHIFT	Shift	Shift length	JD Logistics / recruitment based sorting sample	Recruitment/job-posting level	Recruitment evidence: 4h, 6h, 10h, 12h shifts and night/two shift patterns appear in postings.	Public recruitment information - verification required
JD_EFFECTIVE_TIME	Productivity	Effective working time ratio	JD Logistics / manual sorting	Model assumption	N/A (no public worker level effective working time ratio).	Model assumption
JD_WRONG_SORT	Quality	Manual wrong-sorting rate	JD Logistics / manual sorting	Model assumption	N/A (no public worker level wrong-sorting rate).	Model assumption
JD_MISSCAN	Quality	Manual missed-scan rate	JD Logistics / manual sorting	Model assumption	N/A (no public worker level missed-scan rate).	Model assumption
JD_DAMAGE	Quality	Damage rate	JD Logistics / manual sorting	Model assumption	N/A (no public worker level damage rate).	Model assumption

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ID	Category	Variable	Carrier/context	Scope	Public evidence	Data status
JD_REWORK	Quality	Rework rate	JD Logistics / manual sorting	Model assumption	N/A (no public worker level rework rate).	Model assumption
JD_PEAK	Peak	Peak period human productivity loss	JD Logistics / manual sorting	Model assumption	N/A (no public worker level peak productivity loss).	Model assumption
JD_NIGHT	Night shift	Night shift productivity loss	JD Logistics / manual sorting	Model assumption	N/A (no public worker level night shift productivity loss).	Model assumption

Table B3: Indicator level numerical inputs

Low, base and high values are scenario inputs. Public evidence and assumption status are reported in Table B2.

ID	Low	Base	High	Unit	Source
SF_MANUAL_PPH	300	500	600	parcels/hour/worker	Sohu / maker community repost of Qizhong Intelligence
SF_AUTO_HUB_PPH	71,000	71,000	71,000	parcels/hour/system	Sohu / maker community repost of Qizhong Intelligence
SF_AUTOMATION_INVEST	12.5	12.5	12.5	% efficiency improvement	SF Holding 2025 interim investor presentation
IND_HZ_DAILY	2,200,000	2,200,000	2,200,000	parcels/day/site	CIMC Pteris / CIMC Deli logistics automation case
IND_HZ_ACC	99.99	99.99	99.99	%	CIMC Pteris / CIMC Deli logistics automation case
SF_SHIFT	8	10.5	12.5	hours/shift	User-compiled BOSS Zhipin and Liepin recruitment table
SF_EFFECTIVE_TIME	70	80	85	%	Model assumption based on recruitment shift patterns
SF_WRONG_SORT	0.1	0.5	1	%	No public worker level data; model assumption
SF_MISSCAN	0.1	0.5	1	%	No public worker level data; model assumption
SF_DAMAGE	0.05	0.2	0.5	%	No public worker level data; model assumption
SF_REWORK	0.5	1.5	3	%	No public worker level data; model assumption
SF_PEAK	10	15	25	% decline	CIMC Deli industry description plus model assumption
SF_NIGHT	5	10	15	% decline	User-compiled recruitment table plus model assumption
JD_AUTO_KUNSHAN	9,000	9,000	9,000	parcels/hour/system	People's Daily Overseas Edition / People.cn
JD_AUTO_ASIA_DAILY	4,500,000	4,500,000	4,500,000	parcels/day/site	CIMC Group news release
JD_SORT_WALL_PPH	1,800	1,800	1,800	parcels/hour/equipment	Shanghai Securities News / Sohu repost
JD_PICKER_ASSIST	250	250	250	orders/hour/worker	JD Logistics news centre / People's Daily report
JD_SMART_WAREHOUSE	120	120	120	boxes/hour/robot	Science and Technology Daily
JD_WORKER_PPH	250	400	500	parcels/hour/worker	Composite assumption from JD public picking evidence and SF manual benchmark
JD_SHIFT	4	10	12	hours/shift	User-compiled BOSS Zhipin and Liepin recruitment table
JD_EFFECTIVE_TIME	70	80	85	%	Model assumption based on recruitment shift patterns
JD_WRONG_SORT	0.1	0.5	1	%	No public worker level data; model assumption
JD_MISSCAN	0.1	0.5	1	%	No public worker level data; model assumption
JD_DAMAGE	0.05	0.2	0.5	%	No public worker level data; model assumption
JD_REWORK	0.5	1.5	3	%	No public worker level data; model assumption
JD_PEAK	10	15	25	% decline	JD public peak/automation materials plus model assumption
JD_NIGHT	5	10	15	% decline	User-compiled recruitment table plus model assumption

Table B4: Indicator level modelling basis and recommended use

ID	Modelling basis	Recommended use
SF_MANUAL_PPH	Used as a public benchmark, not as an audited internal KPI.	Use 300/500/600 pph as low/base/high scenarios.
SF_AUTO_HUB_PPH	System level benchmark; not directly comparable with one worker or one humanoid robot.	Use as automation upper bound context; do not compare with a single humanoid robot.
SF_AUTOMATION_INVEST	Used for automation context and seasonal-capacity assumptions.	Use for background and peak capacity assumptions.
IND_HZ_DAILY	Used as industry level benchmark only.	Use for automation maturity and background (peak scenario).
IND_HZ_ACC	Used as industry level benchmark only.	Use as an optimistic accuracy benchmark for robot scenarios.
SF_SHIFT	Used for shift and night-work assumptions; not used as direct productivity evidence.	Use 10-12 hours for formal workers; model part-time roles separately.
SF_EFFECTIVE_TIME	No public worker level evidence; retained as a scenario parameter and marked as N/A in public evidence.	Use 70/80/85% as low/base/high scenarios.
SF_WRONG_SORT	No public worker level evidence; retained as a scenario parameter and marked as N/A in public evidence.	Use 0.1/0.5/1.0% for sensitivity analysis.
SF_MISSCAN	No public worker level evidence; retained as a scenario parameter and marked as N/A in public evidence.	Use 0.1/0.5/1.0% for sensitivity analysis.
SF_DAMAGE	No public worker level evidence; retained as a scenario parameter and marked as N/A in public evidence.	Use 0.05/0.2/0.5% for sensitivity analysis.
SF_REWORK	No public worker level evidence; retained as a scenario parameter and marked as N/A in public evidence.	Use 0.5/1.5/3.0% for sensitivity analysis.
SF_PEAK	No public worker level evidence; retained as a scenario parameter and marked as N/A in public evidence.	Multiply human productivity by (1 - peak loss) in peak scenarios.
SF_NIGHT	No public worker level evidence; retained as a scenario parameter and marked as N/A in public evidence.	Multiply human productivity by (1 - night-shift loss) in night scenarios.
JD_AUTO_KUNSHAN	System level benchmark; not directly comparable with one worker or one humanoid robot.	Use for automation-replacement context; not a single-robot value.
JD_AUTO_ASIA_DAILY	System level benchmark; not directly comparable with one worker or one humanoid robot.	Use as JD automation upper-bound and peak-capacity context.
JD_SORT_WALL_PPH	Equipment level benchmark; not a worker level observation.	Use as comparison for small/standard parcel automation.
JD_PICKER_ASSIST	Warehouse picking benchmark; used only as indirect evidence.	Use as warehouse picking evidence.
JD_SMART_WAREHOUSE	System level benchmark; not directly comparable with one worker or one humanoid robot.	Use as human-robot and robotic-system efficiency context.
JD_WORKER_PPH	No public worker level evidence; retained as a scenario parameter and marked as N/A in public evidence.	Use 250/400/500 pph as low/base/high scenarios.
JD_SHIFT	Used for shift and night-work assumptions.	Use 10-12 hours for formal workers; model 4-6 hour part-time shifts separately.
JD_EFFECTIVE_TIME	No public worker level evidence; retained as a scenario parameter and marked as N/A in public evidence.	Use 70/80/85% as low/base/high scenarios.

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ID	Modelling basis	Recommended use
JD_WRONG_SORT	No public worker level evidence; retained as a scenario parameter and marked as N/A in public evidence.	Use 0.1/0.5/1.0% for sensitivity analysis.
JD_MISSCAN	No public worker level evidence; retained as a scenario parameter and marked as N/A in public evidence.	Use 0.1/0.5/1.0% for sensitivity analysis.
JD_DAMAGE	No public worker level evidence; retained as a scenario parameter and marked as N/A in public evidence.	Use 0.05/0.2/0.5% for sensitivity analysis.
JD_REWORK	No public worker level evidence; retained as a scenario parameter and marked as N/A in public evidence.	Use 0.5/1.5/3.0% for sensitivity analysis.
JD_PEAK	No public worker level evidence; retained as a scenario parameter and marked as N/A in public evidence.	Multiply human productivity by (1 - peak loss) in peak scenarios.
JD_NIGHT	No public worker level evidence; retained as a scenario parameter and marked as N/A in public evidence.	Multiply human productivity by (1 - night-shift loss) in night scenarios.

Table B5: Parcel type manual productivity assumptions

Parcel type	Carrier	Low	Base	High	Unit	Status
Documents	SF Express	450	600	700	parcels/hour/worker	Model assumption - field validation required
Small standard parcels	SF Express	400	550	650	parcels/hour/worker	Model assumption - field validation required
Standard cartons	SF Express	300	450	550	parcels/hour/worker	Model assumption - field validation required
Soft parcels	SF Express	220	350	450	parcels/hour/worker	Model assumption - field validation required
Irregular parcels	SF Express	120	220	320	parcels/hour/worker	Model assumption - field validation required
Heavy/large parcels	SF Express	80	150	250	parcels/hour/worker	Model assumption - field validation required
Documents	JD Logistics	405	540	630	parcels/hour/worker	Model assumption - field validation required
Small standard parcels	JD Logistics	360	495	585	parcels/hour/worker	Model assumption - field validation required
Standard cartons	JD Logistics	270	405	495	parcels/hour/worker	Model assumption - field validation required
Soft parcels	JD Logistics	198	315	405	parcels/hour/worker	Model assumption - field validation required
Irregular parcels	JD Logistics	108	198	288	parcels/hour/worker	Model assumption - field validation required
Heavy/large parcels	JD Logistics	72	135	225	parcels/hour/worker	Model assumption - field validation required

Table B6: Parcel type robotic suitability assumptions

Parcel type	Carrier	Robot suitability	Task coverage	Error-risk multiplier	Basis	Source/notes
Documents	SF Express	High	0.85	0.7	Documents are light and regular; they are expected to have high manual and automation productivity.	https://ir.sf-express.com/media/lokf2io2/2025-interim-investor-presentation-c.pdf
Small standard parcels	SF Express	Very high	0.8	0.8	Small parcels are relatively standardised and frequently appear in automation cases.	https://ir.sf-express.com/media/lokf2io2/2025-interim-investor-presentation-c.pdf ; https://www.sohu.com/a/685639813_120988576
Standard cartons	SF Express	High	0.65	1	Cartons are regular but vary in size and weight; they are suitable as the main model scenario.	Model assumption
Soft parcels	SF Express	Medium	0.4	1.3	Soft parcels deform and create unstable label and grasping positions.	Model assumption
Irregular parcels	SF Express	Medium-low	0.25	1.8	Irregular parcels often require extra review or specialised handling.	https://ir.sf-express.com/media/lokf2io2/2025-interim-investor-presentation-c.pdf
Heavy/large parcels	SF Express	Medium-low	0.2	2	Heavy and large parcels reduce manual productivity and increase damage or safety risk.	Author-compiled dataset of publicly available recruitment listings from BOSS Zhipin and Liepin.
Documents	JD Logistics	High	0.85	0.7	Documents are light and regular; they are expected to have high manual and automation productivity.	https://ir.sf-express.com/media/lokf2io2/2025-interim-investor-presentation-c.pdf
Small standard parcels	JD Logistics	Very high	0.8	0.8	Small parcels are relatively standardised and frequently appear in automation cases.	https://ir.sf-express.com/media/lokf2io2/2025-interim-investor-presentation-c.pdf ; https://www.sohu.com/a/685639813_120988576
Standard cartons	JD Logistics	High	0.65	1	Cartons are regular but vary in size and weight; they are suitable as the main model scenario.	Model assumption
Soft parcels	JD Logistics	Medium	0.4	1.3	Soft parcels deform and create unstable label and grasping positions.	Model assumption
Irregular parcels	JD Logistics	Medium-low	0.25	1.8	Irregular parcels often require extra review or specialised handling.	https://ir.sf-express.com/media/lokf2io2/2025-interim-investor-presentation-c.pdf
Heavy/large parcels	JD Logistics	Medium-low	0.2	2	Heavy and large parcels reduce manual productivity and increase damage or safety risk.	Author-compiled dataset of publicly available recruitment listings from BOSS Zhipin and Liepin.

Table B7: Scenario level worker productivity inputs

Carrier	Scenario	pph	Shift hours	Effective time	Effective parcels/shift	Basis
SF Express	Conservative	300	8	70.0%	1,071	Fatigue, peak pressure and exception items.
SF Express	Base	500	10.5	80.0%	3,213	Uses the 500 pph public benchmark and recruitment shift evidence.
SF Express	Optimistic	600	12	85.0%	5,230	Skilled workers, standard parcels and non-peak conditions.
JD Logistics	Conservative	250	8	70.0%	892	No public worker level JD sorting value; low case informed by public robot-assisted picking evidence.
JD Logistics	Base	400	10	80.0%	2,448	Recruitment based shift assumption and lower than SF 500 pph benchmark.
JD Logistics	Optimistic	500	12	85.0%	4,360	Standard small parcels, skilled workers and non-peak conditions.

Table B8: Quality and productivity loss factors by scenario

N/A is used in the public evidence table for missing observed worker level data.

Carrier	Scenario	Night loss	Peak loss	Wrong-sort	Missed-scan	Damage	Rework
SF Express	Conservative	15.0%	25.0%	1.0%	1.0%	0.5%	3.0%
SF Express	Base	10.0%	15.0%	0.5%	0.5%	0.2%	1.5%
SF Express	Optimistic	5.0%	10.0%	0.1%	0.1%	0.1%	0.5%
JD Logistics	Conservative	15.0%	25.0%	1.0%	1.0%	0.5%	3.0%
JD Logistics	Base	10.0%	15.0%	0.5%	0.5%	0.2%	1.5%
JD Logistics	Optimistic	5.0%	10.0%	0.1%	0.1%	0.1%	0.5%

Table B9: Source overview

Source ID	Source	Supported data	Reliability
S1	Sohu / maker community repost: SF automated sorting system	SF South China hub automatic sorting throughput of 71,000 parcels/hour and a 500 parcels/hour manual comparison benchmark.	Medium
S2	SF Holding 2025 interim investor presentation	Transfer efficiency +12.5%; more than 120 new transfer automation units; more than 380 new AGV forklifts; automated capacity above 5.57 million parcels/day.	High
S3	CIMC Deli Hangzhou Chengnan smart logistics park case	Daily capacity of 2.2 million parcels; sorting accuracy 99.99%; peak promotion volume up to three times normal level.	Medium
S4	People's Daily Overseas Edition / People.cn: JD Kunshan unmanned sorting centre	9,000 parcels/hour; 180 workers saved under comparable site and volume conditions; 10-minute charge for 4 hours of work.	High
S5	CIMC Group: JD Kunshan Asia No.1 phase II	More than 80 automated sorting lines; 10,000 intelligent sorting robots; 99.99% accuracy; more than 4.5 million parcels/day during 618; fivefold efficiency improvement.	Medium-high
S6	Shanghai Securities News / Sohu: JD Logistics automatic sorting wall	Automatic sorting wall: 1,800 parcels/hour; scanning/operation accuracy 99.99%; efficiency improvement 50%.	Medium
S7	JD Logistics news centre / People's Daily report	Robot-assisted picking: about 250 orders/hour per picker; about three times the traditional mode.	Medium-high
S8	Science and Technology Daily: JD Zhi-lang goods-to-person system	One flying-ladder robot handles about 120 boxes/hour; outbound picking efficiency +300%; clothing order level logistics cost -50%.	Medium-high
S9	User-compiled BOSS Zhipin and Liepin recruitment table	Recruitment evidence on shift length, night work, part-time work, 12-hour shifts and wages for SF and JD sorting roles.	Medium
E1	Model assumption	Effective working rate, wrong-sorting, missed scans, damage, rework, night shift loss and peak period loss.	Low-medium

Table B10: Source URLs and use limitations

Source ID	URL or file	Use limitation
S1	https://www.sohu.com/a/206009074_194357	Media repost rather than official SF KPI; used as a public benchmark.
S2	https://ir.sf-express.com/media/lokf2io2/2025-interim-investor-presentation-c.pdf	Company material; useful for automation context, not worker level productivity.
S3	https://cn.cimc-pterisglobal.com/slider/cimc-pteris-showcases-logistics-automation-solutions-in-hangzhou-chengnan/	Company not named; used as an industry benchmark.
S4	https://paper.people.com.cn/rmrbbwb/html/2018-05/30/content_1857991.htm	Authoritative media report; system level data only.
S5	https://www.cimc.com/index.php?a=show&c=index&catid=17&id=3142&m=content	Supplier press release; system level data.
S6	https://www.sohu.com/a/685639813_120988576	Equipment/case level data; not equivalent to manual parcel sorting.
S7	https://www.jingdonglogistics.com/cn/news/200	Warehouse picking order data; not parcel sorting throughput.
S8	https://www.stdaily.com/web/gdxw/2025-08/01/content_379195.html	Warehouse picking system data, not manual parcel sorting.
S9	Author-compiled dataset of publicly available recruitment listings from BOSS Zhipin and Liepin.	Public recruitment extracts; postings change over time and require verification.
E1	No direct public worker level data found	Must be reported as assumptions and replaced with company logs or field observations when available.